VIT | Vidyalankar Institute of Technology
(Accredited A+ by NAAC)
(Autonomous Institute Affiliated to University of Mumbai)

Mid Semester Examination

Branch	Date	Sem.	Roll No. / Exam Seat No.	Subject	Student's Signature	Junior Supervisor's Name and Sign

Question No.	A	B	C	D	E	F	G	H	Total	Total out of (20 /30 / 40)
1										
2										
3										
4										

C Graphics - Sem 5
2025-26
MSE 1
Solution

Examiners Signature	Student's Sign (After receiving the assessed answer sheet)

Q1 Compare and contrast Flood & Boundary fill Algorithm

① Flood fill Algorithm	Boundary Fill Algorithm
→ in flood fill algorithm area is defined with multiple colour	→ in Boundary fill algorithm area is defined with single color
→ The interior is colored with any color	interior point is replaced with new color.
→ May use unpredictable amount of memory to finish, does not subfills swapped	→ More complicated but uses algorithm doesn't req recursion

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5 It is time consuming | It is less time Consuming

Basis for comparision	Flood Fill Algorithm	Boundary fill Algorithm
Basic	It can have area containing several colours	It is defines area with single colour.
Painting proces	Random colour used to colour interior.	by continously searching boundary colour.
Memory consumption	High.	Low.
Speed	Compartively Slower	Fast
Algorithm complexity	Simple Relative	Complicated

Q1 B) Explain Aliasing, Anti aliasing pre and post filtering techniques.

In computer, jagged stair step appearance of lines and edges in rasterized images especially noticable on curves and diagonals.

Anti techniques

Smooth jagged edges, creating appealing realistic image. Pre filtering (area sampling) Post filtering (Super sampling).

Pixel phasing.

Pre filtering (Area Sampling)

This method treats each pixel as amount of object area overlaps pixel area. gray

Post filtering (super sampling)

Super sampling renders image higher resolution than display resolution

Post filtering (Super sampling Reads at higher resolution.

Pixel Phasing (key difference.

Involves slightly offsetting positions. to reduce jaggedness of edges.

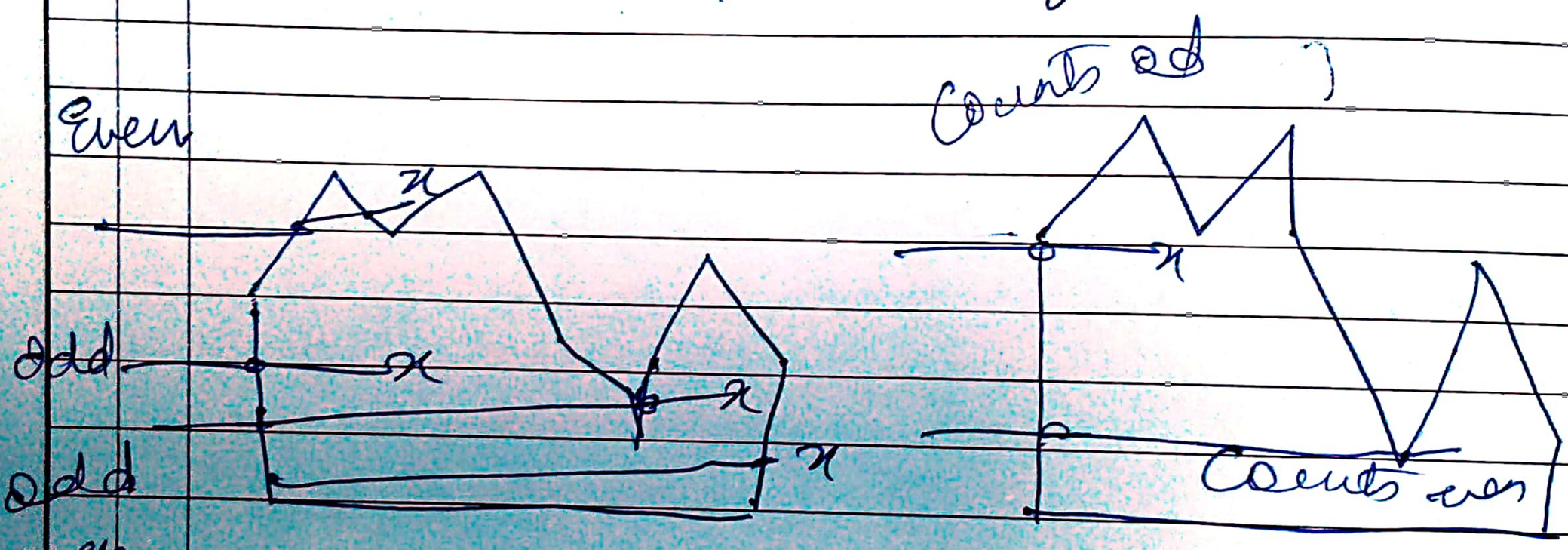
Pixel shifts pixel positions slightly



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Q 4) c (2) Given polygon with set of vertex coordinates apply inside outside test using even odd Rule to determine Polygon lies or not?

→ Inside Outside test using Even Odd Rule determine point lies of this polygon drawing ray from point to infinity. ray intersects polygon edges



Apply Even Odd Rule

① Choose a point

Select point within polygon plane

② Draw ray

Draw line segments chosen point to infinity direction.

③ Count intersection

Determine no of times ray intersects polygon edges.

④ Determine inside outside test.

① if no of intersection is odd point is inside polygon.

② if even, point is outside polygon

Important Considerations

Ray & vertex Intersection :-
If ray perfectly aligns with polygon edge or intersects vertex, adjustment to ray's direction or intersection to avoid

Collinear Points

If point lies exactly Polygon edge. considered inside

Convex vs Concave polygon.

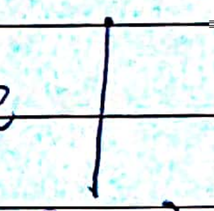
If Even odd rule

8.1) B3 Give detailed Midpoint circle drawing numerical example..

→ (0,0)
R = 10 radius.

→ ① $x_k = 0$ $y_k = 10$ (0,10)

② $d_0 = 1 - R = 1 - 10 = -9$



③ $d_0 < 0$

→ $x_{k+1} = x_k + 1 = 1$

$y_{k+1} = y_k = 10$

$d_{k+1} = d_k + 2x_k + 3$

$= -6$ ie < 0

④ $x_{k+1} \geq y_{k+1}$ at nowstop.

$d_k \rightarrow$	-9	-6	-1	6	-3	8
$d_{k+1} \rightarrow$	-6	-1	6	-3	8	5 ≥ 0

$(x_{k+1}, y_{k+1}) \rightarrow$ (1,10) (2,10) (3,10) (4,9) (5,9) (6,8)

① ② ③ ④ ⑤ ⑥

$x_{k+1} = x_k + 1 = 7$ $d_{k+1} = d_k + 2x_k - 2y_k + 5$

$y_{k+1} = y_k - 1 = 7$

$d_{k+1} = d_k + 2x_k + 2y_k + 5$.

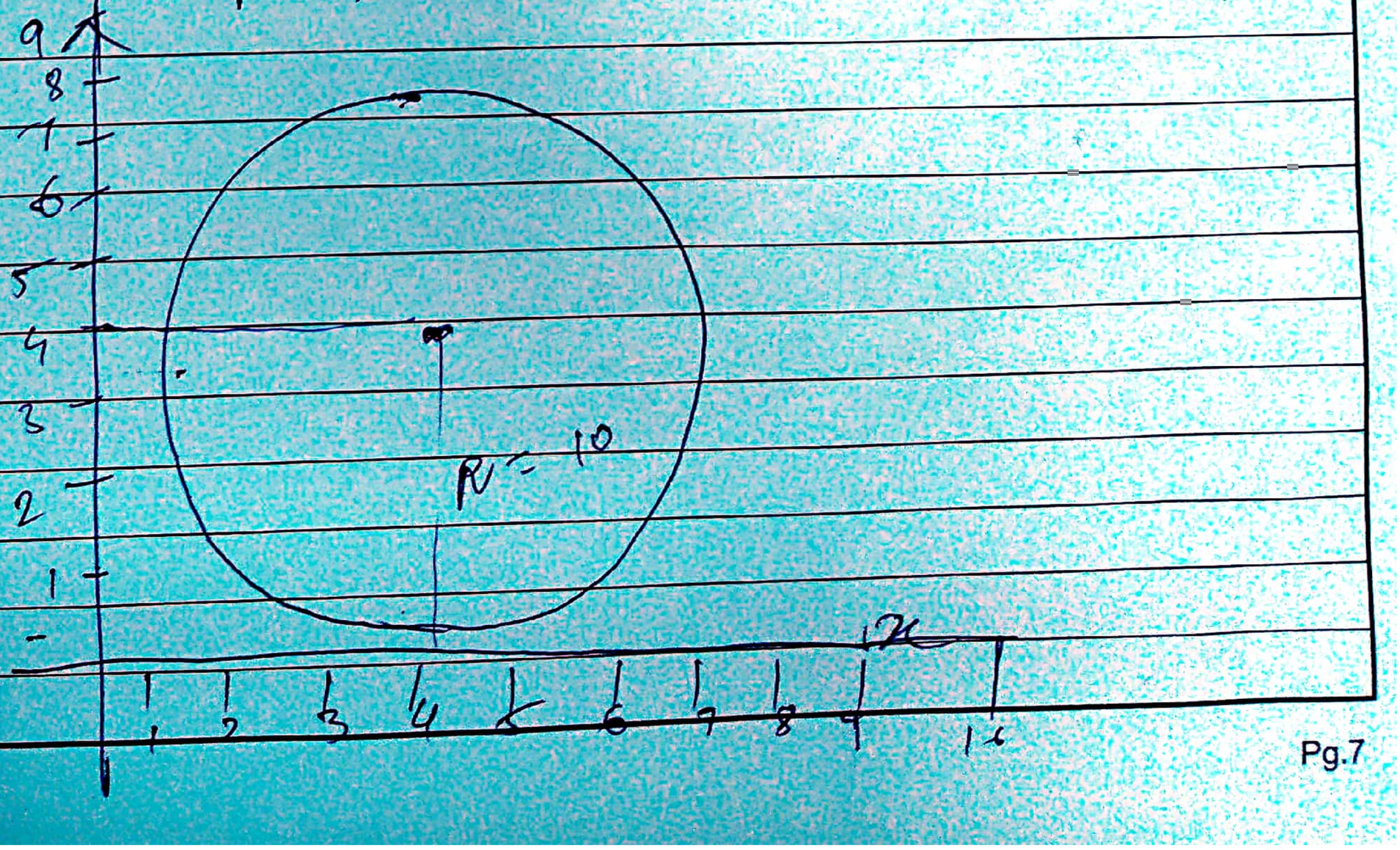
$x_k = y_k$

1: octate

(0, 10)

	Q1	Q2	Q3	Q4
R	(x, y)	(-x, y)	(-x, -y)	(x, -y)
(0, 10)	(0, 10)	(0, 10)	(0, -10)	(0, -10)
	(1, 10)	(-1, 10)	(-1, -10)	(1, -10)
	(2, 10)	(-2, 10)	(-2, -10)	(2, -10)
	(3, 10)	(-3, 10)	(-3, -10)	(3, -10)
	(4, 9)	(-4, 9)	(-4, -9)	(4, -9)
	(5, 9)	(-5, 9)	(-5, -9)	(5, -9)
	(6, 8)	(-6, 8)	(-6, -8)	(6, -8)
	(8, 6)	(-8, 6)	(-8, -6)	(8, -6)
	(9, 5)	(-9, 5)	(-9, -5)	(9, -5)
	(9, 4)	(-9, 4)	(-9, -4)	(9, -4)
	(10, 3)	(-10, 3)	(-10, -3)	(10, -3)
	(10, 2)	(-10, 2)	(-10, -2)	(10, -2)
	(10, 1)	(-10, 1)	(-10, -1)	(10, -1)
	(10, 0)	(-10, 0)	(-10, 0)	(10, 0)

With the help of these points drew circle.



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Q2 (A)

	DDA Line Algorithm	Bresenham Line Algorithm
(1)	Digital differential analyzer	(1) no full form
(2)	Less efficient than Bresenham line alg	while it is more efficient than DDA
(3)	calculation speed of DDA algorithm less than Bresenham.	while calculation speed of Bresenham faster than DDA Alg.
(4)	Costlier than Bresenham line drawing alg.	more precision & accuracy.
(5)	complexity of calculation more complex	complexity of calculation simple
(6)	optimization not provided	optimization provided

→ points on line using Bresenham Line Drawing Algorithm for endpoints,

(9, 18) (14, 22)

$\Delta x = 14 - 9 = 5$

$\Delta y = 22 - 18 = 4$

$P_0 = 2\Delta y - \Delta x = 2(4) - 5 = 3$

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Start point $(x,y) = (9,18)$

Initialize decision parameter (P_0)

$P_0 = 2\Delta y - \Delta x = 3$

$(x,y) = (9,18)$

Iteration 1:

$P_0 = 3 > 0$ inc both x & y.

$x = 9+1 = 10$ $y = 19$ Plot $(10,19)$

$P_1 = P_0 + 2\Delta y - 2\Delta x = 1$

Iteration 2

$P_1 = 1 > 0$ inc both x & y

$x = 11$ $y = 20$ Plot $(11,20)$

$P = P_2 = P_1 + 2\Delta y - 2\Delta x = -1$

if 3

$P_2 = -1 < 0$ inc only x $x = 12$ $y = 20$,

Plot $(12,20)$

$P_3 = P_2 + 2\Delta y = 7$

if 4

$P_3 = 7 > 0$ inc both x & y

$x = 13$ $y = 21$ Plot $(13,21)$.

update $P_4 = P_3 + 2\Delta y - 2\Delta x = 5$

if 5 $P_4 = 5 > 0$ inc x & y

$x = 14$ $y = 22$

$(9,18)$ $(10,19)$ $(11,20)$ $(12,20)$

$(13,21)$ $(14,22)$

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